Hide and SST: Detecting and Analyzing Server-Side Tracking Across the Web

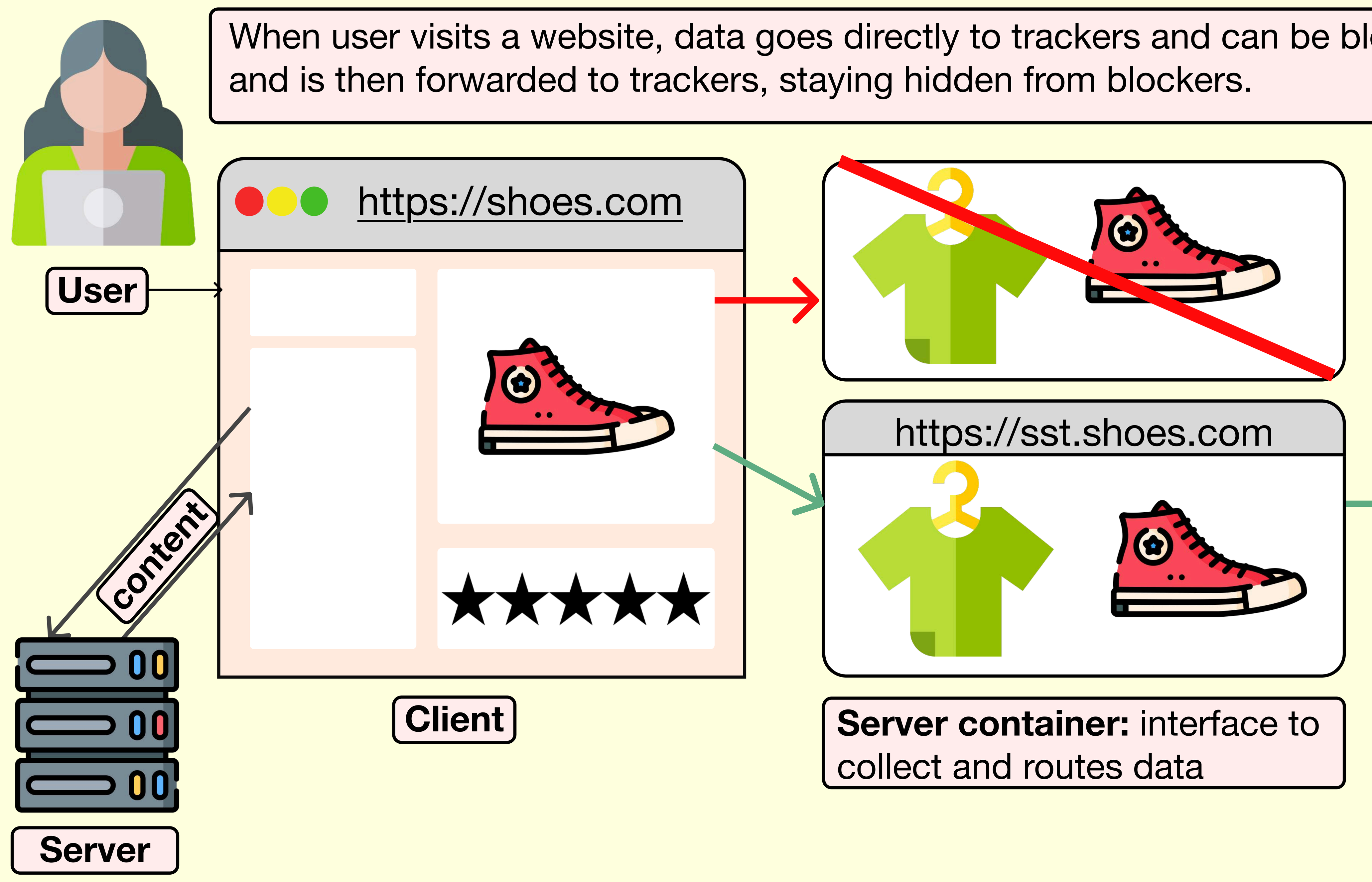
Shambhavi Mittal
shmittal@ucsd.edu

Alisha Ukani
aukani@ucsd.edu

Alex C. Snoeren
snoeren@cs.ucsd.edu

Background and Problem Statement

When user visits a website, data goes directly to trackers and can be blocked. With SST, data goes to a server and is then forwarded to trackers, staying hidden from blockers.



Traditional tracking, i.e. **client-side tracking** happens **in browser** and is **blocked** by privacy tools.

Server-side tracking (SST) hides from users by moving **tracking behind a server.**

Research Question: To help **users not lose visibility** into who collects their data, we look at the presence of **Google's SST in the Tranco List top 10000 websites.**

Trackers

Methodology

Crawl websites

Get call stack for each request

Call stack:
- https://sst.shoes.com/.../collect?...G-...&gtm=...&sst.rnd=...
- https://www.googletagmanager.com/gtm.js/
- https://shoes.com

For each request, if the path contains *r"collect$"*, remove requests with known tracking services in domain.

1. Filter for Google's tracking services.
2. From filtered, remove requests with mentions of known tracking services.

https://sst.shoes.com/g/healthz

Crawl filtered requests by appending */g/healthz/* to URL path prefixes, and keep URLs that respond successfully.

Divide URLs into different groups based on **4 signals.**

Signal 1: the request URL content only contains *"ok"*

https://sst.shoes.com/.../g/healthz
ok

Signal 2: the path of the request URL contains contains *r"collect$"*

https://sst.shoes.com/.../**collect?...**

Signal 3: the request URL contains *r'G-[A-Z0-9]{10}.*gtm='*, *to check for Google ID.*

https://sst.shoes.com/...?...**G-XXXXXXXXXX...&gtm=...**

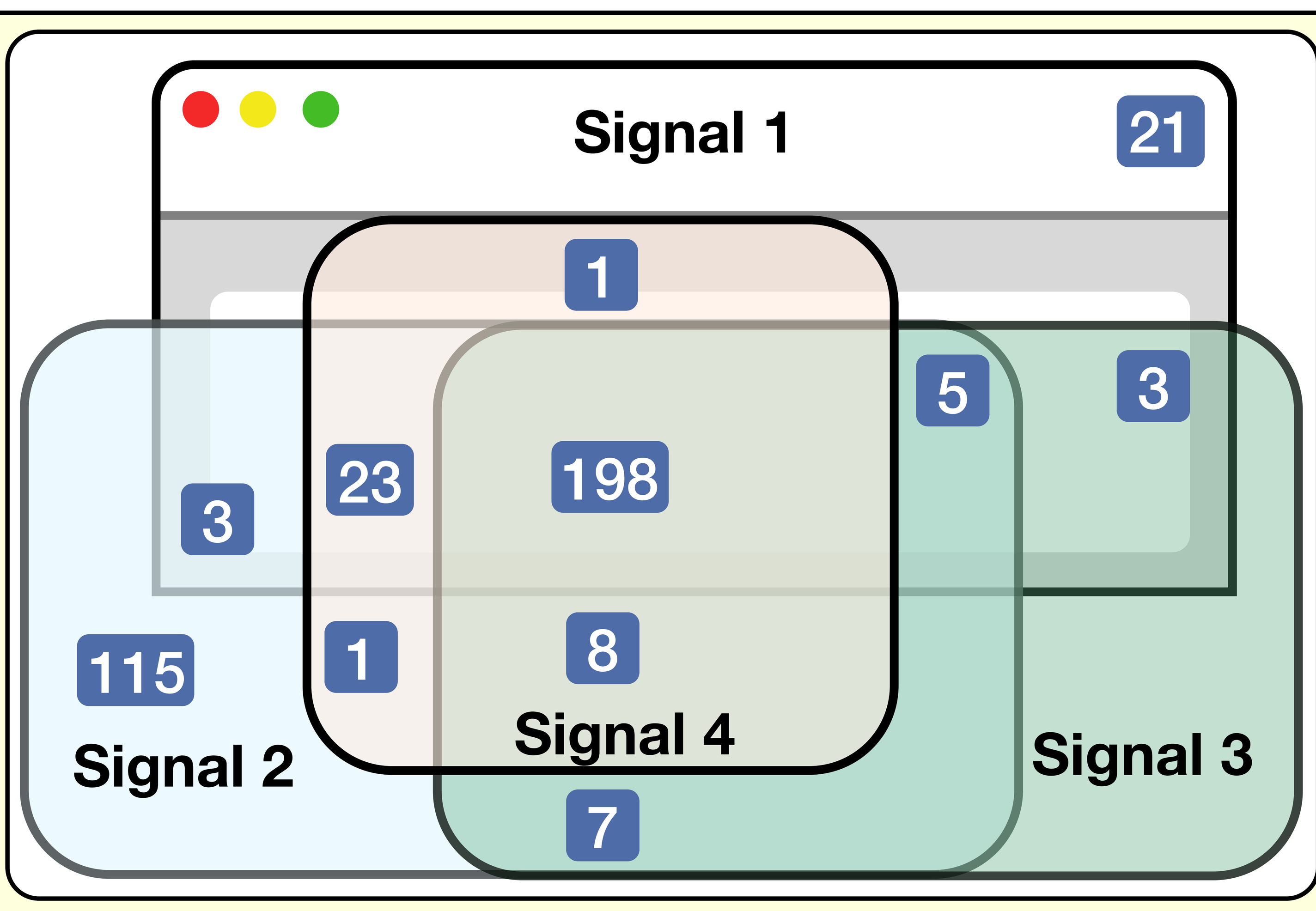
Signal 4: the query of the request URL contains r'sst\.', *which checks for SST variables.*

https://sst.shoes.com/...?**&sst.'**

Signal 3 and 4 are clear, unique indicators of Google's SST. Signal 1 and 2 are validated by the presence of a server container URL within JavaScript files of the website.

Results

- Of 10,000 websites, substantial data was collected from **7697 websites.**
- Identified a total of **385 potential server-side trackers,** each associated with a unique website.
- **292 confirmed server-side trackers** were validated:



- **246** confirmed through presence of either **Signal 3 or Signal 4**
- Additional **46** confirmed using JavaScript files to detect server container URL for the specific website.

Conclusions

- **3.79%** of successfully crawled websites implement **Google's SST,** indicating an active shift by companies to **obscure their tracking practices, threatening user privacy.**
- Because **identifying patterns in request URLs** is effective in large scale, the methodology could help build **better blocking systems to protect users** and **promote more transparency.**

Future Work

- **Ongoing work:** identify more uncaught Google Server-side trackers, while keeping the approach simple and effective.
- **Generalize methodology:** extend to catch SST by other platforms.